

☒ 1. Evidence Checkout Log

The **Evidence Checkout Log** is a documented record used to track who accessed forensic evidence, when, why, and under what circumstances. It ensures:

- **Accountability** of everyone handling the evidence
- **Chain of custody** is preserved
- **Audit trail** is complete for court/legal use

Key fields:

- Evidence ID
- Person checking out/checking in
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Date and time

- Purpose
- Signatures

☒ 2. Handling Evidence

Proper **handling** ensures evidence remains **unaltered and admissible in court**. Key practices include:

- Using **gloves and write blockers**
- Labeling and sealing evidence
- Avoiding **powering on** devices without imaging

- Documenting **every interaction**
- **Hashing** before and after transfer

Handling mistakes can lead to evidence being thrown out in court.

☒ 3. First Response

This is the **initial reaction** when an incident or digital crime is discovered. The **first responder** (IT staff, admin, or investigator) must:

- Secure the scene
- Avoid altering the state of devices
- Photograph and document the scene

- Prevent any tampering
- Report to the Incident Response Team

The **Golden Rule**: Do not power off, plug in, or alter systems unless necessary.

☒ 4. Formulate/Execute Response Strategy

A **response strategy** is the planned approach to contain and investigate an incident.

Formulation involves:

- Understanding the incident scope
- Identifying involved systems
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Defining goals (e.g., containment, evidence collection)

Execution involves:

- Isolating affected devices/networks
- Notifying stakeholders
- Starting preservation and imaging
- Deploying forensic tools

A good strategy ensures minimal data loss and preserves evidence integrity.

☒ 5. Forensic Duplication

A **forensic duplicate (image)** is a **bit-by-bit copy** of storage media, used to analyze data without touching the original.

Features:

- Created using tools like **FTK Imager, dd, Guymager**
- Accompanied by **hash values (MD5/SHA-256)**
- Stored in formats like **.E01, .AFF, .RAW**

Only **duplicates** are analyzed to avoid modifying original evidence.

☒ 6. Authenticate the Evidence

Authentication is proving that the evidence is **original and unaltered**.

Methods:

- **Hashing** (before and after handling)
- **Chain of custody records**
- **Metadata verification**
- **Showing forensic imaging logs**

In court, experts must testify to **how evidence was preserved and verified**.

🔍 7. Investigation

The core part of digital forensics:

Involves:

- Analyzing logs, file systems, registry, emails, network traffic
- Recovering deleted files and timelines
- Connecting evidence to **actions and individuals**
- Drawing **conclusions based on data**

All actions must be logged and repeatable.

⌘ 8. Common Mistakes

Mistakes can **invalidate the evidence**. Examples:

- Powering off a live system without memory capture

- Failing to use write blockers
- Not hashing before/after acquisition
- Incomplete chain of custody
- Overwriting logs or temporary files
- Mishandling mobile/cloud evidence

Proper training and SOPs help avoid them.

☒ 9. Detection

Detection is identifying that an incident or crime has occurred.

Detection methods:

- SIEM tools (e.g., Splunk, ArcSight)
- Alerts from IDS/IPS systems
- Unusual behavior (slow systems, logins at odd times)
- User reports
- Log file monitoring

Early detection reduces damage and improves evidence quality.